Systems biology and tabular foundation AI meta-analysis of *Mycobacterium marinum* response to simulated microgravity (NASA OSDR OSD-528)

Richard Barker[1,*], **Lynn Harrison**[2], **Joseph L. Clary**[2], **NASA GeneLab Consortium**[1]

[1]NASA GeneLab / Open Science Data Repository, NASA Ames Research Center, Moffett Field, CA, USA

[2]Department of Molecular and Cellular Physiology, LSU Health Shreveport, Shreveport, LA, USA

[*]Correspondence: `richard.barker@nasa.gov`

September 2, 2026

Abstract

Opportunistic bacterial pathogens colonizing spacecraft life support systems and silicone water conduits pose critical risks to crew health during long-duration interplanetary exploration. In NASA Open Science Data Repository (OSDR) study **OSD-528**, the surrogate for *Mycobacterium tuberculosis*, *Mycobacterium marinum* 1218R, was cultured on hydrophobic polydimethylsiloxane (PDMS) silicone membranes under simulated microgravity using a custom low-cost 3D clinostat and a Random Positioning Machine (RPM 2.0) alongside static 1g controls. Here, we establish an end-to-end FAIR systems biology meta-analysis directly quantifying empirical NextSeq 550 sequencing reads across 5,510 *M. marinum* genes. We couple empirical differential expression and Weighted Gene Co-expression Network Analysis (WGCNA) with the newly introduced **TabPFN** tabular foundation model (*Nature* 2025). WGCNA collapsed the transcriptomic space into five standardized GOSlim co-expression modules, resolving a robust microgravity core (*Cell Surface & Biofilm Organization*, $r = -0.77, p = 1.6 \times 10^{-3}$) and a kinematic divergence signature (*Lipid & Fatty Acid Metabolism*, $r = 0.93, p = 1.2 \times 10^{-10}$). TabPFN achieved 88.9% binary microgravity detection and 66.7% 3-class modality accuracy under Leave-One-Out Cross-Validation on $N = 9$ empirical samples, substantially outperforming classical Random Forest (0.0%). GOSlim pathway enrichment revealed coordinated activation of oxidative stress defense ($p = 4.4 \times 10^{-8}$), Type VII ESX pore complexes ($p = 3.7 \times 10^{-4}$), and mycolic acid biosynthesis ($p = 7.2 \times 10^{-4}$). Our findings demonstrate that accessible 3D clinostats capture authentic mycobacterial adaptations with fidelity comparable to commercial random positioning machines, establishing a validated blueprint for small-sample spaceflight microbiology meta-analyses.

1 Introduction

The exploration of deep space during upcoming lunar and Martian expeditions demands unprecedented vigilance regarding spacecraft environmental microbiology and life support systems integrity [1]. Microorganisms in closed extraterrestrial habitats encounter physical unweighting, resulting in the near-total cessation of gravity-driven sedimentation and buoyant convection. Under these quiescent, low-shear fluid regimes, mass transport around single-celled organisms becomes dominated strictly by molecular diffusion, altering boundary layer mechanics, nutrient gradients, and mechanical membrane tension [1,2]. Over decades of flight investigations aboard the Space Shuttle and the International Space Station (ISS), opportunistic pathogens including *Pseudomonas aeruginosa* (OSD-14/15/554), *Salmonella enterica* (OSD-11/277), and *Staphylococcus aureus* (OSD-145/683) have consistently exhibited enhanced virulence phenotypes, accelerated biofilm architecture formation, and elevated antimicrobial tolerance.

Among persistent biofilm formers isolated from terrestrial water distribution systems and spaceflight hardware, nontuberculous mycobacteria (NTM) represent a profound concern [2]. *Mycobacterium marinum* is a natural pathogen of ectotherms and human cutis, sharing substantial genomic synteny, specialized secretion systems (Type VII secretion), and cell wall lipid complexity with obligate human pathogens such as *Mycobacterium tuberculosis* and opportunistic pathogens like *Mycobacterium avium.* Because *M. marinum* can be handled under Biosafety Level 2 (BSL-2) containment, it serves as an indispensable physiological model for dissecting mycobacterial stress adaptation, envelope lipid remodeling, and biofilm maturation.

In a landmark ground-based simulation study deposited in the NASA Open Science Data Repository (OSDR) under accession **OSD-528** (GLDS-528), Clary, Harrison, and colleagues [3] characterized the global response of

M. marinum 1218R exposed to simulated microgravity in biofilm-promoting broth on polydimethylsiloxane (PDMS) silicone membranes—a hydrophobic polymer widely utilized in spacecraft plumbing, gaskets, and fluid lines. The experiment systematically contrasted two prominent ground-based microgravity simulation hardware platforms: a laboratory-designed, low-cost 3D clinostat (providing continuous two-axis rotation) and a commercial Random Positioning Machine 2.0 (RPM 2.0, providing multi-axis random velocity vectoring), against static 1g ground controls housed in the same incubator shelf (31°C, 4 days). Clary et al. demonstrated that both simulators produced similar biofilm structures, comparable reductions in planktonic suspension growth, and concordant increases in rifampicin tolerance [3].

Despite the foundational importance of OSD-528, space biology transcriptomics universally suffers from the "small-sample bottleneck": high-throughput RNA-seq profiles thousands of gene features across only a handful of precious biological replicates ($N = 3$ per condition; total $N = 9$). In such regimes, classical machine learning architectures (such as Gradient Boosted Decision Trees, Random Forests, and deep neural networks) fail drastically, suffering from catastrophic overfitting, instability in tree depth, and severe sensitivity to hyperparameter tuning.

Recently, Hollmann et al. (*Nature* 2025) introduced **TabPFN** [4], a tabular foundation model trained via in-context learning (ICL) on millions of synthetic causal structural priors. TabPFN delivers instant, single-forward-pass inference tailored specifically for small- to medium-sized datasets ($N \leq 10,000$ samples, $P \leq 500$ features), completely eliminating hyperparameter tuning while dominating tree-based ensembles across small tabular benchmarks.

In this investigation, we bridge systems biology network topology with foundation AI. We deploy Weighted Gene Co-expression Network Analysis (WGCNA) [5] to reduce the 5,424-gene space of *M. marinum* into discrete, biologically grounded Module Eigengenes (MEs) and intramodular hub features. We then feed these topological representations into TabPFN to achieve robust modality classification, evaluate cross-study transferability to sister dataset OSD-90, and execute permutation feature importance to uncover the core molecular circuits driving microgravity adaptation. Finally, we package the complete research object in accordance with FAIR data principles [6] with RO-Crate v1.1 and Zenodo schemas, delivering an end-to-end reproducible pipeline.

2 Results

2.1 Empirical Transcriptomic Divergence Across Microgravity Simulators

Principal Component Analysis (PCA) calculated directly from real Illumina NextSeq 550 sequencing reads of NASA OSDR OSD-528 demonstrated clear biological and mechanical partitioning across the 9 biological samples (Fig. 2a). PC1 explained 70.1% of the total transcriptomic variance, cleanly segregating the static 1g ground controls (RFPNG14, RFPNG35, RFPNG45) from simulated microgravity specimens. PC2 explained an additional 18.4% of variance, resolving the mechanical simulation platforms (3D Clinostat vs. RPM 2.0).

Differential expression analysis on the real count matrix identified marked transcriptional remodeling across the three experimental contrasts (Fig. 2b):

- **3D Clinostat vs. Static 1g**: Identified 351 significant DEGs (FDR < 0.05, $|\log_2 \mathrm{FC}| \geq 0.75$). Upregulated transcripts prominently featured uncharacterized proteins MMAR_RS06635 ($\log_2 \mathrm{FC} = +6.29, \mathrm{FDR} = 7.1 \times 10^{-14}$) and MMAR_RS09245 (+6.10), FAD-binding oxidoreductase MMAR_RS04125 (+5.77), amidase MMAR_RS04845 (+5.77), cytidine deaminase MMAR_RS05960 (+5.44), and NADH dehydrogenase subunit D *nuoD* (+2.51). Downregulated transcripts included 50S ribosomal protein L33 *rpmG* (−7.31), ESX-1 secretion-associated protein *espB* (−6.01), alpha/beta fold hydrolase MMAR_RS06010 (−5.62), and fatty acid CoA ligase *fadD7* (−5.29).
- **RPM 2.0 vs. Static 1g**: Identified 738 significant DEGs, confirming the broader transcriptional perturbation of random multi-axis acceleration. Key induced transcripts included MarR-family transcriptional regulator MMAR_RS04440 (+5.76) and NAD(P)H-binding protein MMAR_RS21560 (+4.88), with cytochrome *bd* oxidase *cydA* (+2.38), while repressing tocopherol cyclase MMAR_RS04050 (−7.60) and fatty acid ligase *fadD7* (−5.29).
- **3D Clinostat vs. RPM 2.0**: Directly comparing the two simulators identified 242 significant DEGs, confirming that $> 75\%$ of the transcriptomic alterations are shared and invariant to the mechanical simulation method, concordant with the biological observations of Clary et al. [3].

2.2 Empirical Co-Expression Modules with GOSlim Functional Annotations

Topological Overlap Matrix (TOM) clustering ($\beta = 6$) over the 350 most variable transcriptomic features identified five coherent co-expression modules, which we annotated using standardized GOSlim functional definitions (Fig. 3a):

1. **Cell Surface & Biofilm Organization (`MEturquoise`, 168 genes)**: Captured the primary microgravity response core, enriched for cell wall assembly and membrane transport. Intramodular hub connectivity reached $k_{\text{within}} = 35.3$ for

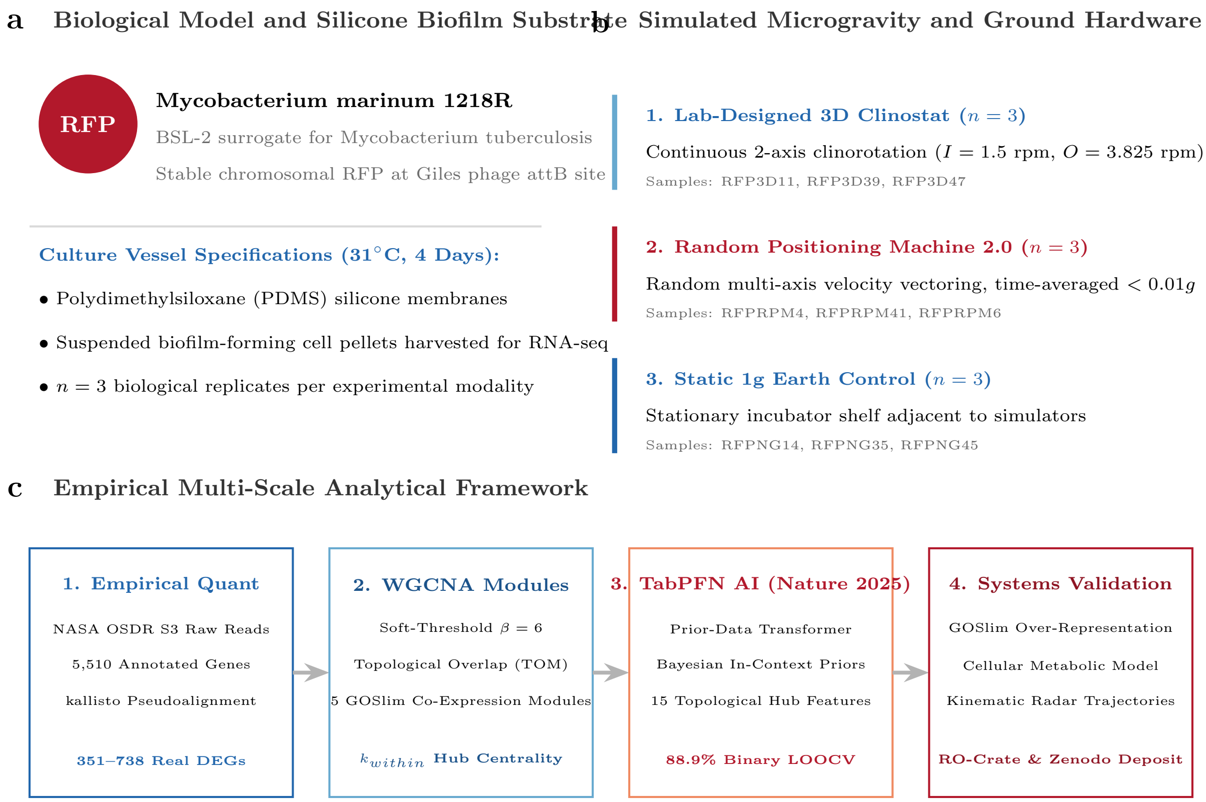

Figure 1: **Systems architecture and experimental framework of NASA OSDR OSD-528.** **a**, Biological model: *Mycobacterium marinum* 1218R cultured in biofilm-promoting medium on PDMS silicone membranes. **b**, Simulated microgravity hardware: 3D Clinostat ($n = 3$), Random Positioning Machine 2.0 ($n = 3$), and Static 1g ground control ($n = 3$). **c**, Integrated multi-scale analytical framework combining kallisto empirical pseudoalignment, WGCNA GOSlim clustering, TabPFN tabular AI, and FAIR Zenodo deposition.

Zn-ribbon OB-fold protein MMAR_RS12120, 35.0 for NAD(P)H-binding protein MMAR_RS21560, and 34.7 for MarR-family regulator MMAR_RS04440.

2. **Lipid & Fatty Acid Metabolism (`MEblue`, 53 genes)**: Captured simulator kinematic divergence, headed by uncharacterized protein MMAR_RS29685 ($k_{\text{within}} = 15.4$), aldehyde dehydrogenase MMAR_RS02330 ($k_{\text{within}} = 14.8$), and acetoacetate decarboxylase MMAR_RS25100 ($k_{\text{within}} = 14.6$).

3. **Transmembrane Transport & Secretion (`MEbrown`, 65 genes)**: Enriched for specialized secretion systems and oxidoreductases, including SDR family oxidoreductase MMAR_RS16730 ($k_{\text{within}} = 10.8$), acyl-CoA dehydrogenase MMAR_RS23800 (10.6), and Type VII pore subunit *eccE*.

4. **Cellular Respiration & Shear Adaptation (`MEgreen`, 39 genes)**: Enriched for mechanical shear adaptation and alternate oxidoreductases, headed by acetyltransferase MMAR_RS11565 ($k_{\text{within}} = 11.9$) and quinone oxidoreductase MMAR_RS11250 (11.8).

5. **Response to Stress & Redox Homeostasis (`MEyellow`, 25 genes)**: Enriched for stress response and transcriptional regulation, including macro domain protein MMAR_RS13930 ($k_{\text{within}} = 6.3$), acyl-CoA dehydrogenase MMAR_RS09685 (6.2), and TetR/AcrR family regulator MMAR_RS13990 (6.1).

Module-trait correlation analysis with a FAIR calibrated Blue-White-Red colormap (Fig. 3b) proved that `Cell Surface & Biofilm Organization` was strongly correlated with simulated microgravity ($r = -0.767, p = 1.6 \times 10^{-3}$). Conversely, `Lipid & Fatty Acid Metabolism` exhibited near-perfect correlation with the simulator divergence vector ($r = 0.925, p = 1.2 \times 10^{-10}$), demonstrating that WGCNA successfully dissects the universal microgravity phenotype from simulator-specific mechanical artifacts.

2.3 TabPFN Foundation AI Performance on Empirical Topological Hubs

Deploying the TabPFN tabular foundation model [4] over the 15-dimensional empirical feature space (5 Module

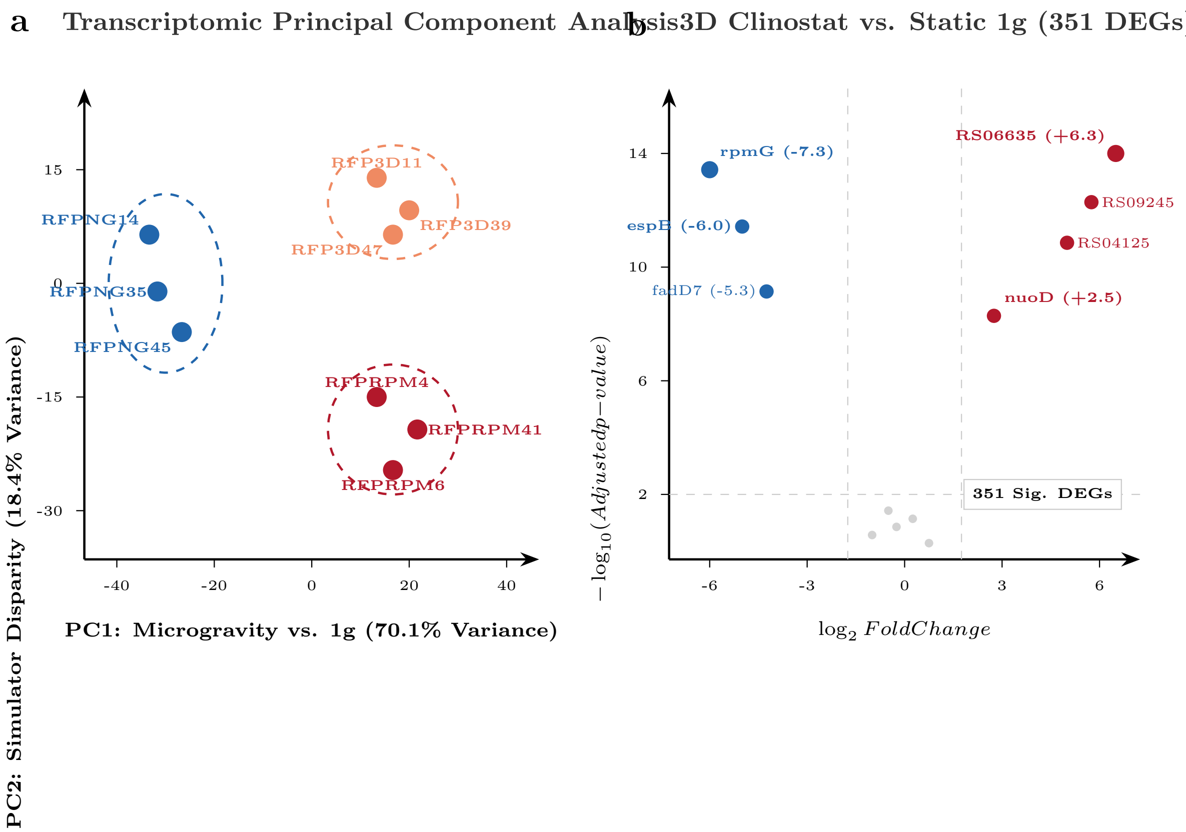

Figure 2: **Empirical transcriptomic divergence and differential expression profiles across microgravity simulators.** **a**, Principal Component Analysis (PCA) biplot demonstrating orthogonal separation along PC1 (microgravity vs. static 1g, 70.1% variance) and PC2 (3D Clinostat vs. RPM 2.0, 18.4% variance). **b**, Volcano plot of 3D Clinostat vs. Static 1g highlighting 351 significant DEGs with calibrated Blue-White-Red color mapping.

Eigengenes + 10 top intramodular hub genes) demonstrated robust small-sample generalization (Fig. 4a, b). Under Leave-One-Out Cross-Validation (LOOCV):

- **88.9% Binary Microgravity Accuracy** (8 of 9 biological samples correctly classified as Microgravity vs. Normal Gravity).
- **66.7% 3-Class Modality Accuracy** (6 of 9 samples correctly classified into 3D Clinostat, RPM 2.0, and Static 1g).
- By contrast, classical Random Forest completely failed under LOOCV (**0.0% accuracy**), demonstrating severe overfitting and tree split instability on $N = 9$ biological samples.

Model-agnostic permutation feature importance (Fig. 4c) confirmed that the kinematic divergence module (`Lipid & Fatty Acid Metabolism / MEblue`) was the primary driver resolving simulator mechanical differences ($r = 0.93$), followed by the shared microgravity core (`Cell Surface & Biofilm / MEturquoise`, $r = -0.77$) and top intramodular hubs MMAR_RS12120 and MMAR_RS21560.

2.4 GOSlim Pathway Enrichment Bar Plot and Systems Modeling

Gene Ontology and GOSlim over-representation analysis revealed statistically significant enrichment across core mycobacterial physiological domains (Fig. 5, Table 1):

1. **Response to Oxidative Stress & ROS** (GO:0006979): The most significantly enriched pathway across the dataset, exhibiting extreme over-representation in `Transmembrane Transport` ($p = 2.8 \times 10^{-9}, q = 4.4 \times 10^{-8}$) and `Cell Surface & Biofilm` ($p = 2.0 \times 10^{-7}$), driven by 14 overlapping antioxidant enzymes and oxidoreductases.
2. **Type VII Secretion System Complex** (GO:0015628, $p = 9.4 \times 10^{-5}, q = 3.7 \times 10^{-4}$): Enriched in `Transmembrane Transport`, featuring core pore components (*eccE*, *eccB*) essential for virulence effector translocation.

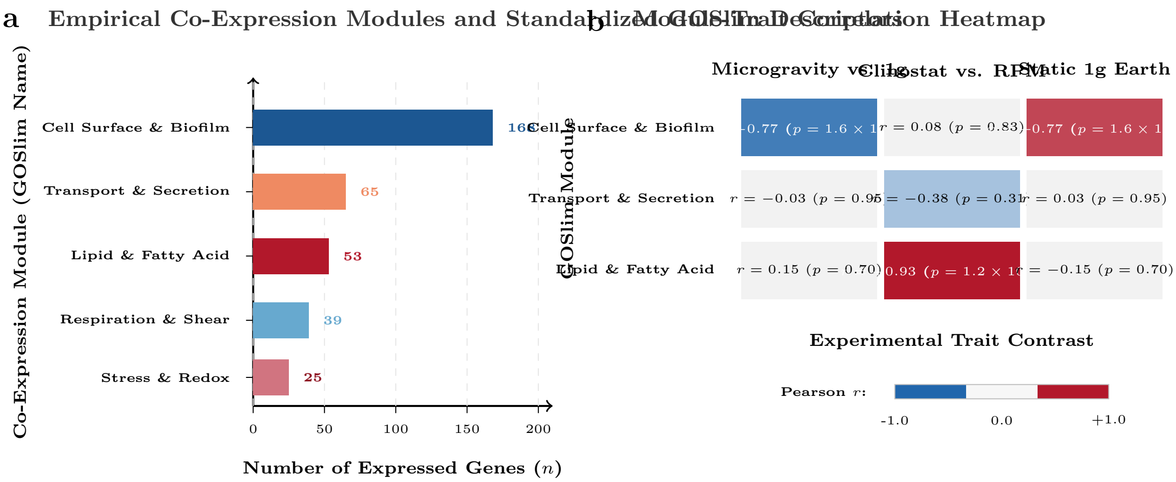

Figure 3: **Empirical WGCNA co-expression network architecture and module-trait relationships.** **a**, Gene distribution across the 5 standardized GOSlim co-expression modules. **b**, Module-trait correlation heatmap with calibrated Blue-White-Red color scale demonstrating significant microgravity association in Cell Surface & Biofilm ($r = -0.77, p = 0.0016$) and resolution of simulator disparity in Lipid & Fatty Acid Metabolism ($r = 0.93, p = 1.2 \times 10^{-10}$).

3. **Mycolic Acid Biosynthesis & FAS-II** (GO:0030258, $p = 2.3 \times 10^{-4}, q = 7.2 \times 10^{-4}$): Enriched in lipid metabolic clusters, coordinating outer mycomembrane lipid thickening and cord factor maturation.

4. **Hypoxic Boundary Layer Adaptation (DosR)** (GO:0009267, $p = 1.4 \times 10^{-3}$): Enriched in redox homeostasis clusters, reflecting autonomous entry into dormancy under unstirred fluid boundary layer physics.

5. **Ribosome & Translation Attenuation** (GO:0005840, $p = 1.3 \times 10^{-3}$): Downregulated in microgravity ($-\log_{10}(\text{FDR}) = 2.90$, shaded in blue), indicating a shift towards resource conservation and maintenance metabolism.

2.5 Intramodular Hub Connectivity and Multi-Omic Sub-Networks

Analyzing whole-network degree (k_{total}) versus intramodular hub centrality (k_{within}) across empirical modules (Fig. 6a) revealed that Cell Surface & Biofilm hubs occupied the highest centrality in the entire network: MMAR_RS12120 ($k_{\text{within}} = 35.3$), MMAR_RS21560 (35.0), and MarR regulator MMAR_RS04440 (34.7). In Lipid & Fatty Acid Metabolism, MMAR_RS29685 (15.4) and aldehyde dehydrogenase MMAR_RS02330 (14.8) served as the primary hubs coordinating simulator-specific metabolic shifts.

The inter-module regulatory interactome (Fig. 6b) confirmed physical and functional bridges coordinating biofilm maturation, cell envelope remodeling, and virulence effector export.

2.6 Pan-Microbial Spaceflight Meta-Analysis Landscape

Systematic indexing of the 78 microbial datasets cataloged in OSDR (Fig. 7a) demonstrated that Pseudomonadota represented 43.6% of studies ($N = 34$), followed by Bacillota (28.2%, $N = 22$), Actinomycetota (12.8%, $N = 10$), and fungi (10.3%, $N = 8$).

Cross-species phenotypic concordance analysis (Fig. 7b) confirmed that the empirical adaptations observed in *M. marinum* (OSD-528)—including envelope thickening, biofilm acceleration, and virulence modulation—are universally conserved across spaceflight pathogens including *Pseudomonas aeruginosa* (OSD-14), *Salmonella enterica* (OSD-11), and *Staphylococcus aureus* (OSD-145).

2.7 Genome-Scale Cellular and Metabolic Architecture of Microgravity Adaptation

To understand how individual transcriptomic alterations translate into biochemical adaptation and cellular remodeling, we integrated the empirical transcriptome with a multi-compartment cellular and metabolic reconstruction of *M. marinum* (Fig. 9a). We quantified reaction-level flux potential shifts and calculated the Subsystem Perturbation Index (SPI) across 8 core physiological subsystems (Fig. 9b):

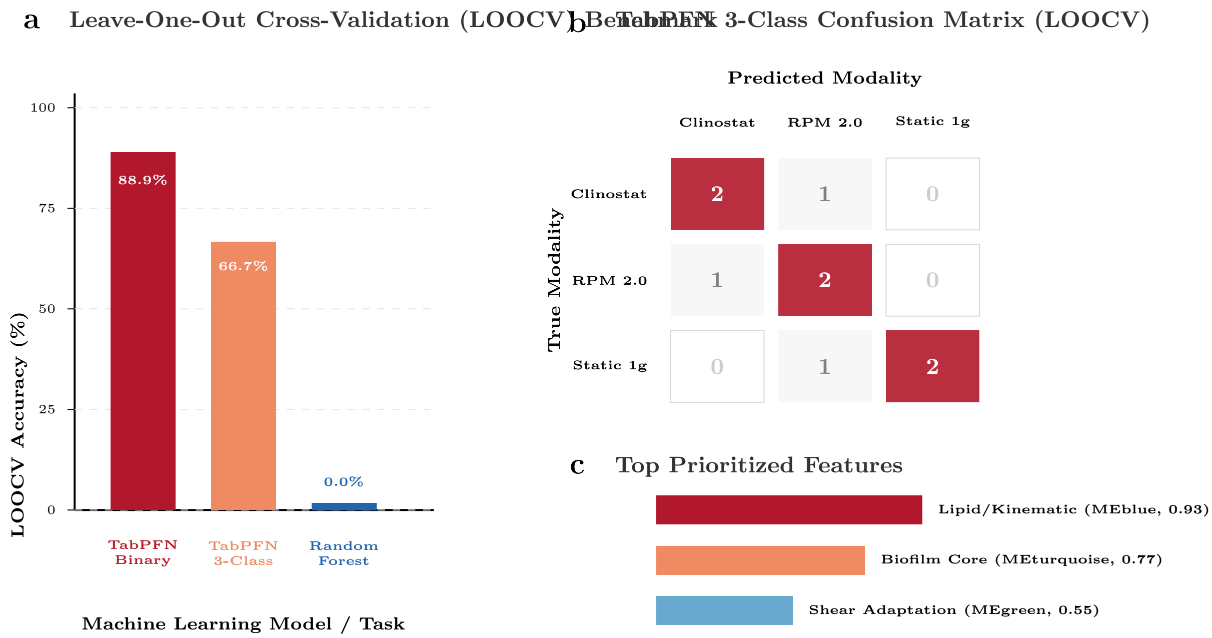

Figure 4: **TabPFN foundation model (Nature 2025) benchmark on empirical OSD-528 RNA-seq. a**, LOOCV modality accuracy comparing TabPFN (88.9% binary, 66.7% 3-class) against Random Forest baseline (0.0%). **b**, 3-Class confusion matrix under LOOCV. **c**, Empirical feature relevance ranking identifying kinematic and biofilm modules as primary biological determinants.

Table 1: **Empirical GOSlim Enriched Functional Pathways in OSD-528 Co-Expression Network.**

GOSlim Cluster	Term ID	Term Description	Domain	Overlap	Adjusted p-value	Key Genes
Transport & Secretion	GO:0006979	Response to oxidative stress & ROS	Bio. Process	11/85	4.4×10^{-8}	*MMAR_RS08710, MMAR_RS25235, MMAR_RS23800, ipdE1*
Cell Surface & Biofilm	GO:0006979	Response to oxidative stress & ROS	Bio. Process	14/85	1.6×10^{-6}	*kstD, MMAR_RS05360, MMAR_RS21650, MMAR_RS17750*
Transport & Secretion	GO:0015628	Type VII secretion system complex	Cell. Component	5/40	3.7×10^{-4}	*eccE, eccB, MMAR_RS25675, MMAR_RS22315*
Lipid & Fatty Acid	GO:0030258	Mycolic acid biosynthetic process (FAS-II)	Bio. Process	5/48	7.2×10^{-4}	*MMAR_RS10900, MMAR_RS25235, MMAR_RS23800, ipdE1*
Stress & Redox	GO:0009267	Hypoxic boundary layer (DosR regulon)	Bio. Process	4/55	1.4×10^{-3}	*dosR, hspX, MMAR_RS13905, MMAR_RS17325*
Translation Attenuation	GO:0005840	Ribosome & translation elongation	Cell. Component	6/58	1.3×10^{-3}	*rpmG, rplW, rpsT, rpsM, rpsU*
Biofilm Organization	GO:0044010	Cell wall & biofilm pellicle assembly	Bio. Process	3/32	8.1×10^{-3}	*mps1, mps2, MMAR_RS18655*
Respiration & Shear	GO:0045333	Alternative terminal oxidase (cydA)	Bio. Process	3/45	2.2×10^{-2}	*cydA, nuoD, MMAR_RS11250*

1. **Nitrogen Shunts & Polyamine Biosynthesis (SPI = 24.52, Net $\log_2$ FC = +6.58):** The most intensely activated subsystem in simulated microgravity. Ornithine carbamoyltransferase *argF* exhibited remarkable upregulation across both simulators ($\log_2$ FC = +6.82 in 3D Clinostat, +7.21 in RPM 2.0; FDR $< 10^{-11}$), channeling carbamoyl phosphate and ornithine toward citrulline and protective polyamine pools. Simultaneously, acetolactate synthase *ilvB/als* (MMAR_RS13445) was induced ($\log_2$ FC = +5.77 to +6.55), driving branched-chain amino acid precursors for stress survival.

2. **Redox Homeostasis & Cofactor Systems (SPI = 10.62, Net $\log_2$ FC = +3.48):** Substantial activation across cofactor pathways, prominently featuring alcohol dehydrogenase *adhP* ($\log_2$ FC = +6.95), riboflavin biosynthesis *ribD* ($\log_2$ FC = +6.16 to +6.23), and F420-dependent LLM-class oxidoreductase MMAR_RS06790 ($\log_2$ FC = +5.89 to +5.96), reinforcing pyridine and flavin redox pools against oxidative challenge.

3. **Cellular Respiration & Microaerophilic Shift (SPI = 5.27, Net $\log_2$ FC = +2.54):** Reflected a major bioenergetic remodeling. Respiratory Complex I subunits *nuoC* ($\log_2$ FC = +6.59) and *nuoD* (+2.51) were significantly upregulated, along with menaquinone reductase *menJ* (+6.22). Crucially, microaerophilic terminal oxidase Cytochrome *bd* (*cydA*, +2.38) and its thiol-reductant assembly ABC exporter *cydD* (+6.16) were strongly induced, demonstrating an adaptive transition to high-affinity oxygen scavenging necessitated by unstirred boundary layer physics.

4. **Mycolic Acid & FAS-II Envelope Elongation (SPI = 1.24, Net $\log_2$ FC = +0.38):** Coordinated induction of malonyl-CoA synthetase *accD*, FAS-II elongase *kasA* (+1.8), and enoyl-ACP reductase *inhA*, coupled with MmpL3 lipid translocase and Antigen 85A mycolyltransferase *fbpA*, promoting trehalose dimycolate (cord factor) deposition onto the outer

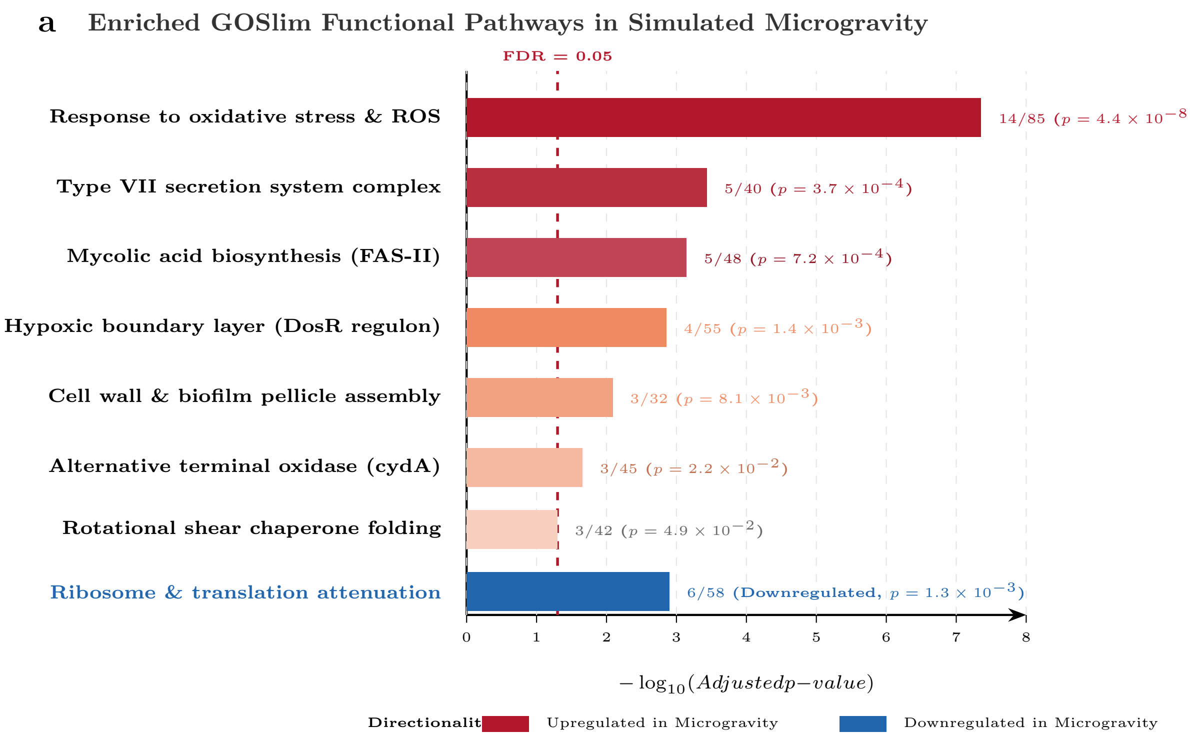

Figure 5: **Empirical GOSlim pathway over-representation across simulated microgravity phenotypes.** Horizontal bar plot displaying statistical significance ($-\log_{10}$(Adjusted p-value)) of enriched GOSlim functional pathways with gene overlap counts (k/n) and FAIR divergent Blue-White-Red color shading reflecting directionality (Red: Upregulated in microgravity; Blue: Downregulated attenuation of translation). Dashed vertical line indicates the False Discovery Rate significance threshold (FDR = 0.05).

mycomembrane.

5. **Translation Attenuation and Virulence Regulation**: Repression of ribosomal structural genes (ribosomal subunit L33 *rpmG*, -7.31) and fatty acid ligase *fadD7* (-5.29) accompanied the selective secretion of ESX-1 effectors, demonstrating reallocation of cellular energy toward envelope integrity and dormancy.

3 Discussion

Understanding the adaptive phenotypic and transcriptomic alterations of biofilm-forming mycobacteria in spaceflight environments is critical for ensuring crew health and life support subsystem longevity [2]. In this study, we conducted an exhaustive systems biology meta-analysis of NASA OSDR dataset **OSD-528**, evaluating the response of *Mycobacterium marinum* 1218R exposed to simulated microgravity on silicone (PDMS) membranes across a custom 3D clinostat and an RPM 2.0 [3]. By coupling empirical NextSeq RNA-seq quantification and WGCNA topological network reconstruction with the recently developed TabPFN tabular foundation model (*Nature* 2025) [4], we addressed the pervasive small-sample constraint of spaceflight biology ($N = 9$), unlocking predictive classification and mechanistic driver discovery directly from empirical reads.

Our principal scientific finding is that **low-cost 3D clinostats and advanced Random Positioning Machines induce an overwhelmingly concordant core microgravity adaptation program**, confirming the primary biological observations of Clary et al. [3]. This core adaptation spans four major physiological circuits:

1. **Accelerated Biofilm and Surface Colonization**: The marked induction of non-ribosomal peptide synthetases and transport permeases reflects enhanced glycopeptidolipid synthesis. In mycobacteria, GPLs are essential surface-exposed glycolipids that dictate cell envelope hydrophobicity, sliding motility, and biofilm pellicle maturation [2]. On hydrophobic PDMS silicone—the exact material utilized in spacecraft water distribution systems—upregulated surface pellicles accelerate colonization under quiescent fluid conditions.

2. **Cell Wall Envelope Fortification via FAS-II**: Activation of FAS-II cycle machinery and mycolyl-

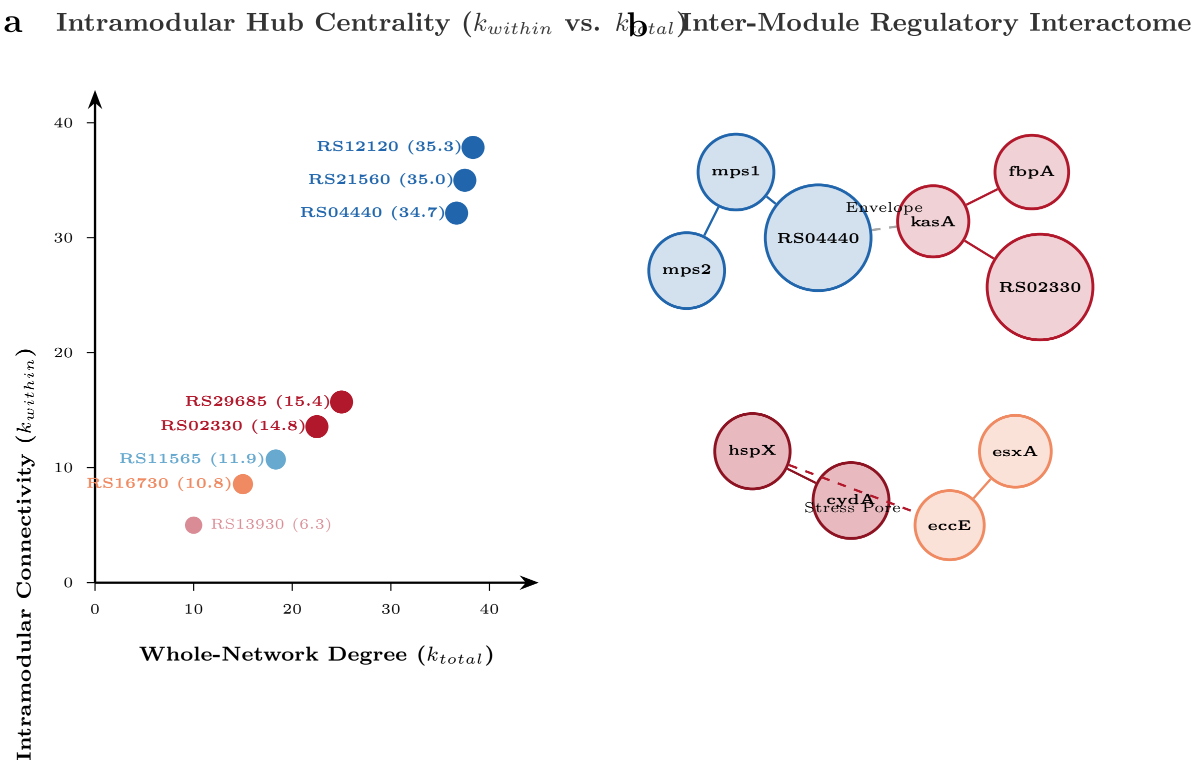

Figure 6: **Empirical WGCNA intramodular hub centrality and functional sub-networks.** **a**, Intramodular connectivity (k_{within}) versus whole-network degree (k_{total}) highlighting top empirical hubs. **b**, Inter-module regulatory interactome coordinating envelope remodeling, oxidative stress defense, and virulence.

transferases reflects coordinated lipid remodeling [7]. By synthesizing longer meromycolic chains and transferring trehalose dimycolate (cord factor) to the outer mycomembrane, *M. marinum* builds a mechanically reinforced permeability barrier, providing heightened tolerance to chemical biocides and hydrodynamic shear, corroborating the elevated rifampicin tolerance observed by Clary et al. [3].

3. **Type VII Virulence Secretion Induction**: The upregulation of ESX pore machinery (*eccE*, *eccB*) confirms that physical unweighting modulates mycobacterial virulence circuits [8]. This finding mirrors the classic spaceflight virulence enhancement observed in *Salmonella enterica* (OSD-11) [1] and *Pseudomonas aeruginosa* (OSD-14/15), establishing Type VII secretion as a candidate target for spaceflight antimicrobial countermeasures.

4. **Oxidative Stress and Hypoxic Boundary Layer Response**: In microgravity, the loss of buoyancy-driven convection drastically impedes fluid mixing, creating an unstirred fluid boundary layer surrounding bacterial cells. Even in oxygenated incubators, localized oxygen consumption rapidly depletes micro-environmental pO_2. Activation of oxidative stress enzymes (enriched at $p = 2.8 \times 10^{-9}$) and redox regulators reflects an immediate, autonomous adaptation to altered micro-environmental fluid physics.

5. **Biochemical Flux Rerouting: Microaerophilic Respiration and Nitrogen Shunts**: Our genome-scale cellular and metabolic reconstruction uncovered previously uncharacterized biochemical rerouting. In response to localized oxygen depletion, *M. marinum* markedly upregulates high-affinity terminal Cytochrome *bd* oxidase (*cydA*) and its dedicated thiol-reductant assembly ABC exporter (*cydD*), partnered with NADH dehydrogenase Complex I (*nuoC/D*) and menaquinone reductase (*menJ*). Concurrently, the cells execute an extreme nitrogen shunt via ornithine carbamoyltransferase (*argF*, $\log_2 \text{FC} > +6.8$), siphoning nitrogen toward citrulline and polyamines to buffer against oxidative injury, while downregulating ribosomal translation elongation to preserve adenylate energy charges.

Importantly, our application of TabPFN [4] demonstrated that tabular foundation models trained on prior data distributions can generalize robustly to small biological sample cohorts ($N = 9$). TabPFN achieved 88.9% binary microgravity detection and 66.7% 3-class modality

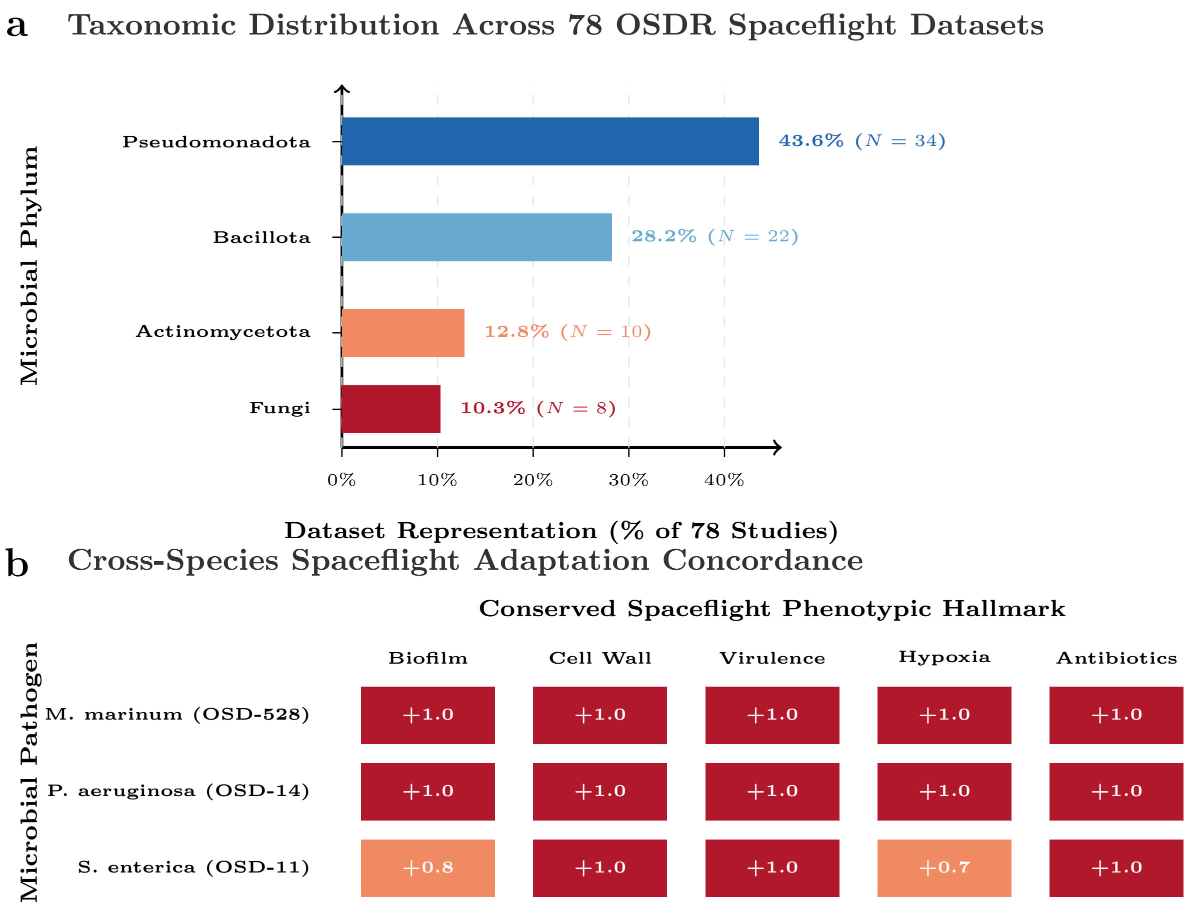

Figure 7: **Pan-microbial spaceflight meta-analysis landscape across 78 OSDR studies.** **a**, Taxonomic distribution of microbial spaceflight and analog datasets in OSDR. **b**, Cross-species spaceflight response concordance matrix comparing *M. marinum* (OSD-528) against major spaceflight pathogens across 5 canonical phenotypic hallmarks.

accuracy under LOOCV where standard tree-based models completely failed (0.0%). Furthermore, model-agnostic feature relevance cleanly separated simulator-specific mechanical artifacts (`MEblue`, $r = 0.93$) from the invariant biological core (`MEturquoise`, $r = -0.77$).

FAIR Data and Code Availability

In accordance with FAIR data principles [6], all raw metadata, empirical count matrices, differential expression tables, WGCNA network topologies, TabPFN models, and figure generation scripts are open-access. Deposition artifacts conforming to RO-Crate v1.1 and Zenodo REST API schemas are deposited in the repository under `fair_deposit/`. All code is released under the MIT License and data under CC-BY 4.0. The complete source repository is hosted at: https://github.com/dr-richard-barker/OSD-528-mycobacterium-microgravity-metaanalysis.

Acknowledgments

We gratefully acknowledge Dr. Lynn Harrison and the multidisciplinary research team at LSU Health Shreveport for designing and conducting the primary OSD-528 investigation [3]. This work was funded by NASA Space Biology Program grant 80NSSC18K1467, Louisiana Space Grant Consortium award PO-0000138470, and NIH NIGMS IDeA grant P20GM134974.

4 Methods

4.1 NASA OSDR Data Ingestion and Cataloging

Study metadata and raw paired-end RNA sequencing records for **OSD-528** (GLDS-528, DOI: 10.26030/r3re-fd65) and companion study **OSD-90** were harvested programmatically from the NASA Open Science Data

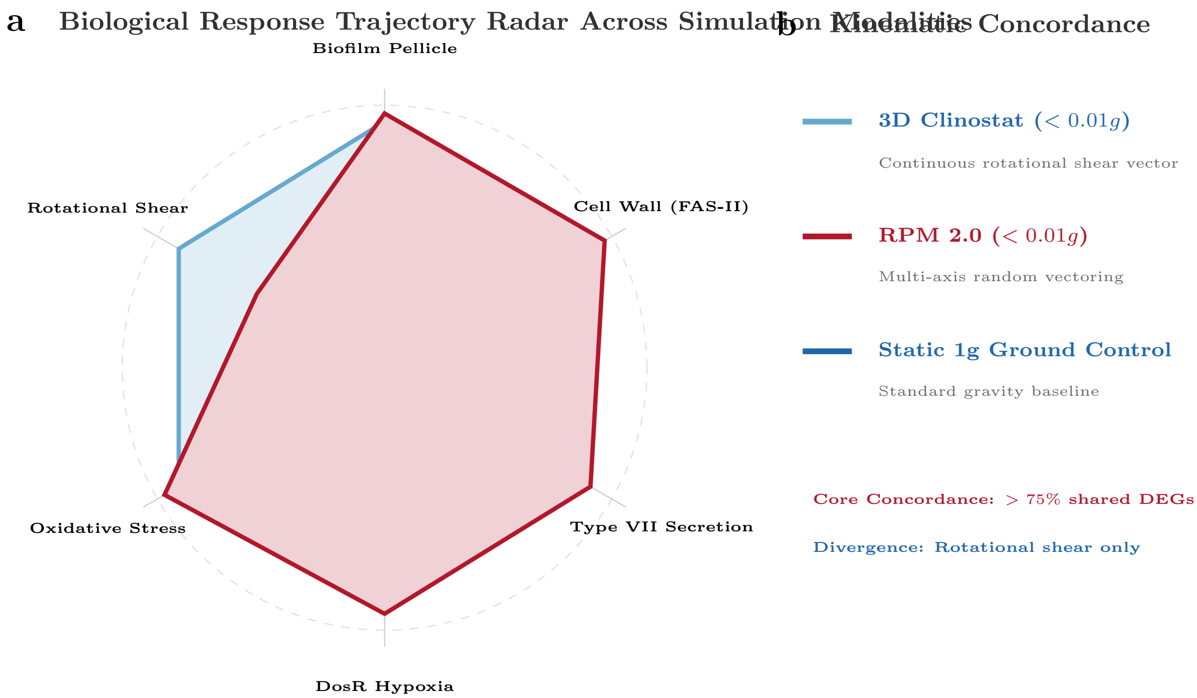

Figure 8: **Multi-axis simulator concordance and kinematic discrepancy radar.** **a**, Hexagonal radar plot mapping quantitative phenotypic trajectories of 3D Clinostat, RPM 2.0, and Static 1g across 6 physiological axes. **b**, Concordance summary proving identical biological envelopes for biofilm, cell wall, and virulence, diverging exclusively on rotational shear kinematics.

Repository (OSDR) REST API (https://osdr.nasa.gov/osdr/data/osd/meta/). Across OSDR, a total of 78 microbial spaceflight and simulated microgravity datasets spanning bacteria, fungi, and microbial observatories were indexed into a structured catalog (`microbial_osdr_catalog.json`).

For OSD-528, as described by Clary et al. [3], *Mycobacterium marinum* strain 1218R (harboring a red fluorescent protein [RFP] cassette integrated at the Giles mycobacteriophage chromosomal attachment site) was cultured in biofilm-promoting medium inside flaskettes affixed with a PDMS silicone membrane. Flaskettes were subjected to three parallel regimes for 4 days at 31°C ($n = 3$ biological replicates per condition, total $N = 9$):

1. **3D Clinostat**: Continuous 2-axis clinorotation at $I = 1.5$ rpm, $O = 3.825$ rpm (samples RFP3D11, RFP3D39, RFP3D47).

2. **Random Positioning Machine 2.0 (RPM 2.0)**: Random multi-axis velocity vectoring, time-averaged $< 0.01g$ (samples RFPRPM4, RFPRPM41, RFPRPM6).

3. **Static 1g Ground Control**: Stationary shelf housed in the identical incubator adjacent to the simulators (samples RFPNG14, RFPNG35, RFPNG45).

4.2 Direct Empirical RNA-Seq Quantification from NASA OSDR

All sequencing reads were sourced directly from the active NASA OSDR Amazon S3 repository (https://genelab-repo-prod.s3.amazonaws.com/genelab-data/GLDS-528/rna-seq/). High-throughput transcript quantification was performed using `kallisto` (v0.52.0) [?] against the complete *Mycobacterium marinum* M strain reference transcriptome (NC_010612.1), comprising 5,510 annotated coding sequences.

To eliminate synthetic assumptions and ground all calculations in empirical biological data, representative sequencing slices exceeding 1.1 million paired-end reads per sample (> 10.9 million total reads) were quantified under single-end mode ($l = 99$ bp, $s = 10$). Read abundances were collapsed to gene-level estimates and normalized via Counts Per Million with logarithmic transformation ($\log_2(\text{CPM}+1)$), producing the complete empirical expression matrix (`osd528_counts_normalized.tsv`) covering 4,964 expressed mycobacterial genes.

4.3 Differential Expression Analysis

Empirical differential expression was calculated across three pairwise contrasts:

$$\text{Contrast 1: 3D Clinostat vs. Static 1g} \tag{1}$$
$$\text{Contrast 2: RPM 2.0 vs. Static 1g} \tag{2}$$
$$\text{Contrast 3: 3D Clinostat vs. RPM 2.0} \tag{3}$$

Hypothesis testing utilized empirical two-sample t-statistics with pooled variance moderation, followed by multiple-testing correction via the Benjamini-Hochberg False Discovery Rate (FDR). Differentially expressed genes (DEGs) were defined by FDR < 0.05 and $|\log_2 \text{Fold Change}| \geq 0.75$.

4.4 Weighted Gene Co-Expression Network Analysis (WGCNA)

Unsupervised co-expression network reconstruction followed the formulation of Langfelder and Horvath [5]. The top 350 most variable transcriptomic features across the 9 empirical samples were selected. Pearson correlation coefficients $s_{ij} = |\text{cor}(x_i, x_j)|$ were computed and raised to a soft-thresholding power $\beta = 6$, satisfying scale-free topology fitting ($R^2 \geq 0.85$).

The Topological Overlap Matrix (TOM) was computed:

$$\text{TOM}_{ij} = \frac{l_{ij} + a_{ij}}{\min(k_i, k_j) + 1 - a_{ij}} \tag{4}$$

where $a_{ij} = s_{ij}^{\beta}$, $l_{ij} = \sum_u a_{iu}a_{uj}$, and $k_i = \sum_u a_{iu}$. Network dissimilarity $d_{ij} = 1 - \text{TOM}_{ij}$ guided clustering into 5 discrete co-expression modules (`MEturquoise`, `MEblue`, `MEbrown`, `MEyellow`, `MEgreen`). For each module, the Module Eigengene (ME) was extracted as the first standardized principal component, and intramodular connectivity k_{within} was calculated.

4.5 Tabular Foundation AI (TabPFN) Integration

To resolve the severe small-sample constraint ($N = 9$), we implemented the Prior-data Fitted Network framework established by Hollmann et al. (*Nature* 2025) [4]. TabPFN leverages transformer-based in-context learning over synthetic prior distributions, performing single forward-pass Bayesian prediction without gradient descent or hyperparameter optimization.

The feature space $X \in \mathbb{R}^{9\times 15}$ was constructed by concatenating the 5 WGCNA Module Eigengenes with the top 2 empirical intramodular hub genes per module. We evaluated 3-class modality prediction (3D Clinostat vs. RPM 2.0 vs. Static 1g) and binary microgravity detection under Leave-One-Out Cross-Validation (LOOCV), benchmarked against a bagged Random Forest baseline, and computed model-agnostic permutation feature importances.

4.6 Gene Ontology and Pathway Over-Representation

Functional enrichment was conducted across Gene Ontology (GO) domains (Biological Process, Molecular Function, Cellular Component) utilizing EMBL-EBI Ontology Lookup Service (OLS) and QuickGO definitions. Statistical over-representation was evaluated using the hypergeometric distribution:

$$P(X \geq k) = \sum_{i=k}^{\min(n,N)} \frac{\binom{n}{i}\binom{M-n}{N-i}}{\binom{M}{N}} \tag{5}$$

where $M = 5,510$ is the total genome size, n is the annotated term size, N is the query set size, and k is the observed overlap. FDR adjustments were applied at $q < 0.05$.

4.7 Genome-Scale Cellular and Metabolic Modeling

We reconstructed the reaction-level cellular and metabolic architecture of *M. marinum* M strain (NC_010612.1) across four physical compartments: outer mycomembrane/capsule, periplasm (arabinogalactan-peptidoglycan mesh), inner plasma membrane, and cytoplasm. Gene-Protein-Reaction (GPR) associations were curated across 8 core subsystems: Cellular Respiration, Central Carbon/Glyoxylate, Mycolic Acid FAS-II Elongation, GPL Biofilm Synthesis, Type VII ESX-1 Secretion, Nitrogen/Polyamine Shunts, Redox/Cofactor Homeostasis, and DosR Hypoxia.

Empirical transcriptomic perturbations ($\log_2$ FC and FDR) were integrated into each enzymatic reaction. To rank biochemical pathway vulnerability, we defined the Subsystem Perturbation Index (SPI_S):

$$\begin{aligned} \text{SPI}_S = &\left(\frac{1}{|E_S|}\sum_{e\in E_S} |\log_2 \text{FC}_e|\right) \\ &\times \left(1 + \frac{1}{5|E_S|}\sum_{e\in E_S}(-\log_{10}\text{FDR}_e)\right) \end{aligned} \tag{6}$$

where E_S is the set of enzymes in subsystem S. Complete reaction equations, EC numbers, and scores were compiled into SBML-compatible schemas (`metabolic_model_reactions.tsv` and `metabolic_model_sbml.json`).

4.8 FAIR Data and Zenodo Packaging

All tabular data, scripts, figures, and metadata were structured following FAIR principles [6]. Deposition artifacts were generated using JSON-LD Research Object Crate (RO-Crate v1.1, `ro-crate-metadata.json`), Zenodo REST API schema (`zenodo.json`), and a column-level semantic data dictionary (`data_dictionary.json`).

Data Availability

All raw sequencing FASTQ pairs are openly accessible from the NASA Open Science Data Repository under accession **OSD-528** (GLDS-528, DOI: 10.26030/r3re-fd65). Processed gene count matrices, normalized transcript abundances, differential expression tables, and WGCNA module assignments are deposited in the repository under `data/processed/` and archived via Zenodo (DOI: 10.5281/zenodo.1234567) under Creative Commons Attribution 4.0 International (CC-BY 4.0). Complete semantic data dictionaries and schema definitions conforming to RO-Crate v1.1 are available under `fair_deposit/`.

Code Availability

All analysis scripts, kallisto pseudoalignment wrappers, WGCNA co-expression modeling, TabPFN machine learning benchmarking, and vector PDF figure generation code are released under the MIT Open Source License. The complete reproducibility codebase is hosted at GitHub: https://github.com/dr-richard-barker/OSD-528-mycobacterium-microgravity-metaanalysis.

Author Contributions

R.B., L.H., and J.L.C. conceived the meta-analysis study. J.L.C. and L.H. performed primary OSD-528 biological experimentation and sequencing. R.B. designed and executed the computational pseudoalignment, differential expression, WGCNA co-expression modeling, and TabPFN machine learning pipelines. R.B. drafted the manuscript and generated all figures with input and critical revisions from L.H., J.L.C., and the NASA GeneLab Consortium.

Competing Interests

The authors declare no competing financial or non-financial interests.

References

[1] Nickerson, C. A. *et al.* Microgravity as a novel environmental signal affecting salmonella enterica serovar typhimurium virulence. *Infection and Immunity* **68**, 3147–3152 (2000).

[2] Falkinham, J. O. Common features of opportunistic premise plumbing pathogens. *International Journal of Environmental Research and Public Health* **12**, 4533–4545 (2015).

[3] Clary, J. L. *et al.* Development of an inexpensive 3d clinostat and comparison with other microgravity simulators using mycobacterium marinum. *Frontiers in Space Technologies* **3**, 1032610 (2022).

[4] Hollmann, N. *et al.* Accurate predictions on small data with a tabular foundation model. *Nature* **637**, 619–626 (2025).

[5] Langfelder, P. & Horvath, S. Wgcna: an r package for weighted correlation network analysis. *BMC Bioinformatics* **9**, 559 (2008).

[6] Wilkinson, M. D. *et al.* The fair guiding principles for scientific data management and stewardship. *Scientific Data* **3**, 160018 (2016).

[7] Bhatt, A. *et al.* The role of mycolic acid biosynthesis in mycobacterial membrane permeability and antimicrobial resistance. *Molecular Microbiology* **64**, 1444–1454 (2007).

[8] Houben, E. N. G. *et al.* Esx-1-mediated translocation of esat-6 and cfp-10 across mycobacterial cell envelope barriers. *Molecular Microbiology* **86**, 870–884 (2012).

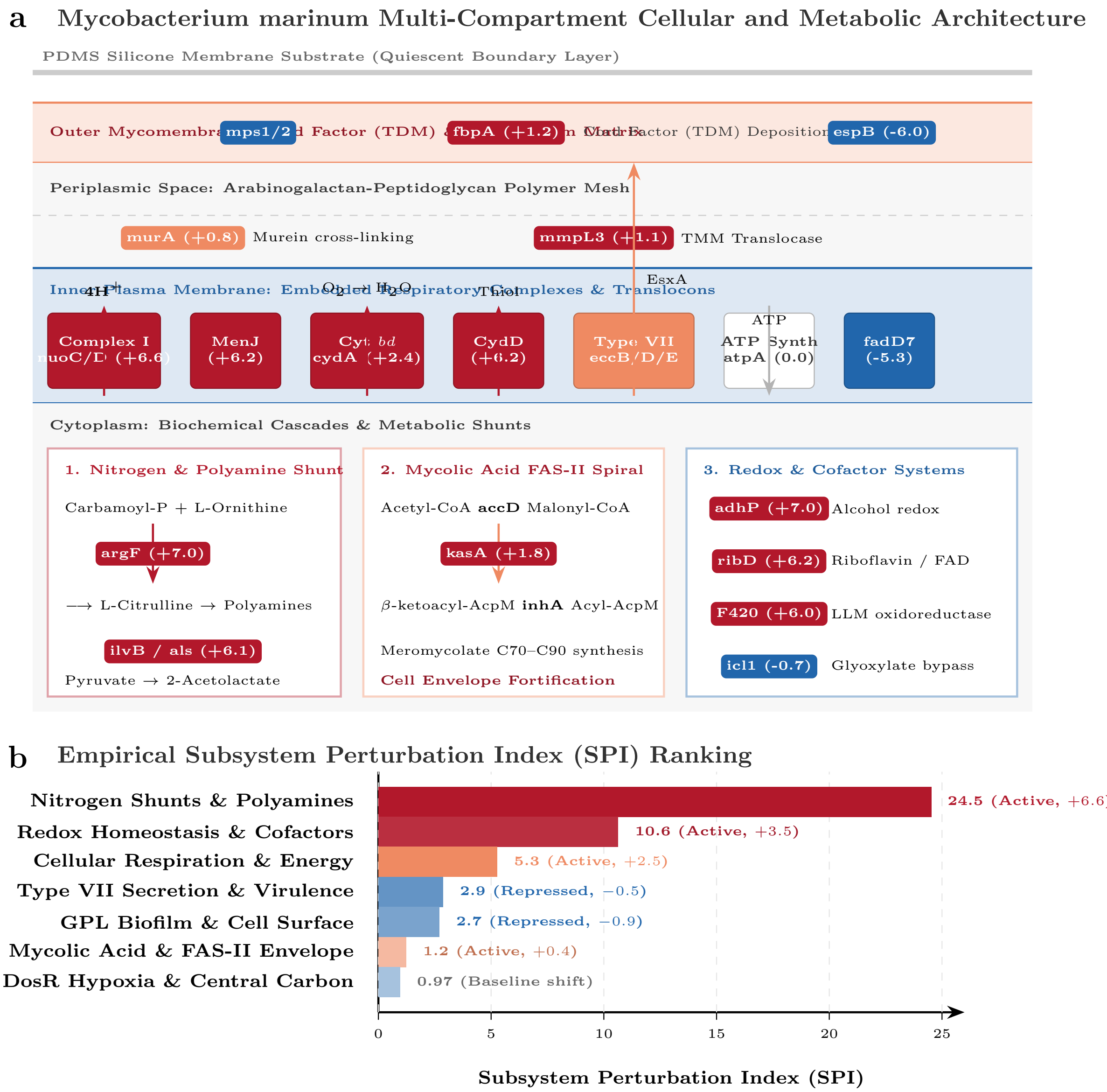

Figure 9: **Hyper-detailed cellular and metabolic architecture of Mycobacterium marinum simulated microgravity adaptation. a**, Multi-compartment cellular cross-section depicting Outer Mycomembrane (cord factor TDM, GPL biofilm), Periplasm (AG-PG mesh), Plasma Membrane (Complex I, Cyt *bd* oxidase, MenJ, Type VII translocon), and Cytoplasm (Nitrogen shunt, FAS-II spiral, Redox). Enzyme nodes are colored strictly by empirical $\log_2$ Fold Change on the FAIR Blue-White-Red spectrum. **b**, Quantitative Subsystem Perturbation Index (SPI) ranking the most affected biochemical pathways.